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Inventor(s)	Kovacs; Tamas

Equalizer clamp assembly, system and method

Abstract

A multi-directional, equalizer clamp device is disclosed, the clamp device being configured to attach and secure objects, such as posts, rails and the like, for a multiplicity of clamping applications that benefit from single-point clamping operation. The clamp device comprises upper and lower body portions which may be configured to clamp to a side rail of a support table, while further configured to receive objects and clamp them along a secondary clamping direction. The clamp device further comprises a control assembly, such as a knob, to facilitate and achieve simultaneous clamping along both primary and secondary clamping directions at a single point of control. A jogging clamp for clamping to variable sized objects in a primary direction comprises a plurality of spherical, indexing recesses disposed in the lower jaw and matching spherical washer disposed on the control assembly configured to allow large, indexed vertical variation in the clamping range.

Inventors:	Kovacs; Tamas (Burlington, CT)
Applicant:	INNOVATIVE MEDICAL PRODUCTS, INC. (Plainville, CT)
Family ID:	1000008778413
Assignee:	INNOVATIVE MEDICAL PRODUCTS, INC. (Plainville, CT)
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Primary Examiner: King; Anita M

Attorney, Agent or Firm: Wasserbauer Law, LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of, claims priority to, and the benefit of, co-pending U.S. Non-Provisional application Ser. No. 17/857,535, filed Jul. 5, 2022, entitled, "SINGLE-ACTION, MULTI-DIRECTIONAL CLAMP ASSEMBLY, SYSTEM AND METHOD", which claims priority to Ser. No. 17/073,334, filed Oct. 17, 2020, granted U.S. Pat. No. 11,400,006 issued on Aug. 2, 2022, entitled, "EQUALIZER CLAMP ASSEMBLY, SYSTEM AND METHOD", which in turn claims priority to 62/916,674, filed Oct. 17, 2019, entitled, "EQUALIZER CLAMP ASSEMBLY, SYSTEM AND METHOD", of which the disclosures are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates to a clamp assembly and, more particularly, to a clamp apparatus, system and method for simultaneously locking multiple components with different orientations.

BACKGROUND OF THE INVENTION

(2) When clamping multiple components with different orientations, conventional clamps use individual forcing mechanisms in order to secure each object to the clamp body. For example, a clamp may have a primary opening to secure to an object (e.g., a rail, rod, bar, standing seam, or the like) in one direction (e.g., x-direction) and other openings to insert other objects disposed in another direction (e.g., y-direction). For the opening(s) in each separate direction, e.g., x or y, a separate mechanism is used to apply force to the clamp body such as, for example, a knob, handle, or other tightening device to urge the parts of the clamp body to tighten and/or loosen around the object. The additional components for each forcing mechanism add complexity and cost to the manufacturing of the clamp assembly. It would be advantageous to reduce the number of components in the clamp assembly as well as to use a single forcing mechanism to operate the clamp for multiple objects, e.g., using a single knob or handle.

(3) Consequently, there is a need for a clamp apparatus, system and method using a fastener and/or forcing mechanism having a single action for locking multiple objects disposed in different numerous directions that achieves clamping in multiple directions simultaneously.

SUMMARY OF THE INVENTION

(4) The present invention provides numerous applications in various industries would benefit from such a clamp assembly. For example, a clamp assembly application to attach objects to the side rail of a surgical support can improve access to the patient for the surgeon. It is to be appreciated by one of skill that the clamping can be arranged on the side rail of a support table in either direction. The invention is described using an upper portion and a lower portion for the body parts of the clamp, which is merely for convenience of illustration and should not be deemed limiting to the invention in any way.

(5) A clamp assembly for surgical procedures further requires sterilization and improved sterilization can be accomplished with a clamp body having open channels that may be cleared by standard sterilization techniques.

(6) Applications also exist in attaching a clamp assembly to various objects and/or structures. For example, a single action fastener that clamps multiple objects can be used advantageously in connection with a clamp assembly for the standing seam of the metal roof, thereby reducing the time of the installer on the roof and/or structure, use of available space, costs of manufacturing, and improving the aesthetic appearance of these objects installed on the structure. As will be appreciated by one skilled in the art, the clamp apparatus, system and method may find further used for other applications to clamp other items and things including mounting to a structure such as a wall, shingled roof or other rooftop, and other structures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a front, top, right, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientations using a single fastener and/or forcing mechanism, according to a first embodiment of the invention;

(2) FIG. 2 illustrates a perspective view of the clamp apparatus of FIG. 1 in the environment of attaching a surgical positioner using elongated posts and/or pins to a side rail of a support table, according to an embodiment of the invention;

(3) FIG. 3A illustrates an exploded, front, top, right, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientation using a single fastener and/or forcing mechanism thereof;

(4) FIG. 3B illustrates an exploded, rear, bottom, right, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientation using a single fastener and/or forcing

mechanism thereof;

- (5) FIG. 4A illustrates a section view of the clamp apparatus of FIG. 1 in a closed position thereof;
- (6) FIG. 4B illustrates a section view of the clamp apparatus of FIG. 1 in an open position thereof;
- (7) FIG. 5 illustrates a front, top, right, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientation using a single fastener and/or forcing mechanism, according to another embodiment of the invention;
- (8) FIG. 6A illustrates an exploded, front, top, left, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientation using a single fastener and/or forcing mechanism thereof;
- (9) FIG. 6B illustrates an exploded, rear, bottom, right, perspective view of a clamp apparatus for clamping multiple objects disposed in different orientation using a single fastener and/or forcing mechanism thereof;
- (10) FIG. 7A illustrates a section view of the clamp apparatus of FIG. 5 in a first clamped position thereof;
- (11) FIG. 7B illustrates a section view of the clamp apparatus of FIG. 5 in a first unclamped position thereof;
- (12) FIG. 7C illustrates a section view of the clamp apparatus of FIG. 5 in a jogging position thereof;
- (13) FIG. 7D illustrates a section view of the clamp apparatus of FIG. 5 in a second unclamped position thereof; and
- (14) FIG. 7E illustrates a section view of the clamp apparatus of FIG. 5 in a second clamped position thereof.

DETAILED DESCRIPTION OF INVENTION(S)

(15) Non-limiting embodiments of the present invention will be described below with reference to the accompanying drawings, wherein like reference numerals represent like elements throughout. While the invention has been described in detail with respect to the preferred embodiments thereof, it will be appreciated that upon reading and understanding of the foregoing, certain variations to the preferred embodiments will become apparent, which variations are nonetheless within the spirit and scope of the invention.

(16) The terms “a” or “an”, as used herein, are defined as one or as more than one. The term “plurality”, as used herein, is defined as two or as more than two. The term “another”, as used herein, is defined as at least a second or more. The terms “including” and/or “having”, as used herein, are defined as comprising (i.e., open language). The term “coupled”, as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically.

(17) Reference throughout this document to “some embodiments”, “one embodiment”, “certain embodiments”, and “an embodiment” or similar terms means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of such phrases or in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments without limitation.

(18) The term “or” as used herein is to be interpreted as an inclusive or meaning any one or any combination. Therefore, “A, B or C” means any of the following: “A; B; C; A and B; A and C; B and C; A, B and C”. An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

(19) The drawings featured in the figures are provided for the purposes of illustrating some embodiments of the present invention, and are not to be considered as limitation thereto. Term “means” preceding a present participle of an operation indicates a desired function for which there is one or more embodiments, i.e., one or more methods, devices, or apparatuses for achieving the desired function and that one skilled in the art could select from these or their equivalent in view of

the disclosure herein and use of the term “means” is not intended to be limiting.

(20) FIGS. **1** through **7A-7E** illustrate certain embodiments and features generally shown as clamp apparatus, system, and method **100**. One or more of the concepts described therein may be useful in combination with the teachings of U.S. Non-Provisional application Ser. No. 17/857,535 and variations and combinations thereof. Consequently, clamp **100** is described generally in the environment of surgical positioners that is non-limiting as clamp **100** may be useful for a variety of applications such as in the construction industry, and other fields of use. Another intended use for clamp **100** may be to secure and hold a work tablet or electronic device, for ease of use and display while in the field. Further examples of intended use may be clamping, either alone or in combination with a device or object, of or to a solar panel, a snow rail system, HVAC equipment, electrical equipment, structural equipment, and/or construction equipment. Therefore, the applications of clamp **100** are also non-limiting.

(21) Referring to FIGS. **1** to **4A-4B** illustrating a first embodiment for a clamp apparatus, system, and method for a single-action clamp capable of locking multiple components disposed in a variety of orientations is generally shown as element **100**. The clamp **100** may comprise four component parts: a clamp assembly **110**, a fastener assembly **150**, a washer assembly **160**, and a control assembly **170**. The clamp assembly **110** may comprise an upper body **111** configured to operatively connect to a lower body **130**. The fastener assembly **150** may comprise a hex bolt **151** with a threaded portion **152** configured to be disposed through fastener openings **123**, **137** of the upper body **111** and lower body **130**, respectively, so as to join using forces applied by a washer assembly **160** and control assembly **170**. The upper body **111** of the clamp assembly **110** comprises the top portion **112**, a bottom portion **113**, a front portion **114**, a back portion **115** portion, and side portion **116** that is designated as individual sides **116a** and **116b**. The upper body **111** may also be considered as including an upper base **111a** and an upper arm **111b**, as shown in FIG. **4A**. The upper body **111** may further comprise one or more openings **117**, designated as openings **117a** and **117b**, shown in FIG. **3B**, and which can be formed between the top portion **112** and the bottom portion **113**, and extend therethrough. As illustrated in FIG. **2**, the openings **117a** and **117b** of FIG. **3B** may be configured to receive one or more posts **104** for attaching, for example, a patient positioner **103** to the side rail **102** of the operating room table **101** according to an embodiment of the invention. The upper body **111** also comprises one or more slot openings **118** designated as slot openings **118a** and **118b** in FIGS. **1** and **3A**. The slot openings **118a** and **118b** are configured to receive one or more posts **104** for attaching a patient positioner **103** to the side rail **102** of the support table **107**, as shown in FIG. **2**, according to an embodiment of the invention. The clamp assembly **110** can be assembled such that the slot openings **118a** and **118b** are located adjacent the slot recesses **138** designated as slot recesses **138a** and **138b** in FIGS. **1** and **3A**, to accomplish single action locking of multiple components in different orientations. The upper body **111** comprises a back portion **115** that can be configured with a upper side rail arm **119** that can further have a side rail register **126** and a side rail lip **127** so as to couple the clamp **100** to the side rail **102**. The upper body **111** also comprises a channel portion **120** having a slot segment **121** and shoulder segment **122** adapted to be operably connected to and receive a key protrusion **140** and corresponding slot segment **141** and shoulder segment **142** that may be formed in the lower body **130** of the clamp assembly **110**.

(22) The upper body **111** comprises one or more fastener openings **123** disposed in an angled portion **114a** of the front portion **114** and extending through to the back portion **115**, so as to receive the fastener assembly **150** therein. Similarly, the lower body **130** comprises one or more fastener head openings **137**, such that operably connecting the channel portion **120** and the key protrusion **140** disposes the one or more fastener openings **123** and the one or more fastener head openings **137** in aligned relationship, such that the fastener assembly **150** may extend through the fastener opening **123**. The fastener head opening may have one or more rotational inhibitors **137a** that interfere with a rotational motion of the hex bolt **151**. Furthermore, the washer assembly **160**

may then be disposed on the threaded portion **152** of the hex bolt **151** of the fastener assembly **150** extending therethrough, such that the control assembly **170** may be secured thereto, as shown in FIGS. **1-4B**. The fastener opening **123** may be in the form of a slot allowing for vertical translation of the washer **160** and control **170** assemblies during tightening of the control knob **171**. Although the bolt **151** is represented herein as having a hexagonal head **151a**, the shape of the bolt head **151a** may take any shape other than hexagonal, including but not limited to a head having at least one protrusion that is sufficient to maintain a constant rotational position of the bolt head **151a** with respect to, e.g., the lower body **130**. For example, the bolt head **151a** may be square shaped or triangular shaped such that as control assembly **170** is tightened or loosened, the bolt head **151a** remains in the same rotational position.

(23) In operation, tightening control assembly **170** forces the upper body **111** and the lower body **130** toward one another to achieve clamping forces in multiple directions simultaneously. In a first securing operation, a clamping force may be achieved in a direction that allows upper and lower bodies **111** and **130** to fixedly attach to, for example, a rail **102**. In a second securing operation, a clamping force may be achieved in a direction that allows upper and lower bodies **111** and **130** to fixedly attach to, for example, a post **104**. The shapes are non-limiting thereto that are secured by the upper and lower portions as contemplated herein as items or components may take any shape, e.g., circular, square, rectangular. Consequently, equivalent structures may be used to connect to other items or components so that these may be affixed via clamp **100** and are allowed to move in multiple directions regarding the first and second directions.

(24) The lower body **130** comprises a top portion **131**, a bottom portion **132**, a front portion **133**, a back portion **134**, and side portion **135** that is designated as individual sides **135a** and **135b**. The lower body **130** may also be considered as including a lower base **130a** and a lower arm **130b**, as shown in FIG. **4A**. The top portion **131** and front portion **133** comprise a key protrusion **140** extending therefrom. The key protrusion **140** comprises a slot segment **141** and a shoulder segment **142**. The lower body **130** of the clamp body assembly has a lower side rail arm **136** extending from the top portion **131** that may have a similar side rail register **126** and side rail lip **127** for ease of registering on the side rail **102** of a support table **101**. The support table **101** may be configured with a tabletop or upper table surface **107**. In one embodiment, the lower body **130** further comprises one or more slot recess **138**, designated as individual slot recess **138a** and **138b**, so as to form points of connection, along with slot openings **118a** and **118b**, configured to receive posts **104**. The lower body **130** further includes one or more fastener head openings **137** receiving the hex bolt **151** of the fastener assembly **150** that can be located and formed by a milled slot extending partially in the lower body **130** between the front portion **133** and the back portion **134**. In this manner, the clamp assembly **110** can be assembled such that the slot openings **118a** and **118b** can be located adjacent respective slot recesses, **138a** and **138b**, to accomplish single action locking. By tightening the control assembly **170**, which advances washer assembly **160** along threaded portion **152** of the fastener assembly **150** toward hex bolt **151**, the clamp **100** achieves a clamping force where, for example, the upper body **111** and lower body **130** are forced in two directions simultaneously.

(25) FIGS. **4A-4B** illustrate the alignment of the fastener **150**, washer **160** and control assemblies **170** when the clamp assembly **100** is in a clamped or loosely clamped position. FIG. **4A** shows the assembled arrangement of the components of the control assembly **170**, including a knob **171** which is fitted with a threaded insert **172** whose threads mate with the fastener threads **152**. The threaded end **152** of the fastener **151** is further coupled to a threaded stop **173**, which is assembled to the fastener **151** through an opening in the knob **171**. Referring to FIG. **4B**, the threaded stop **173** is of a diameter larger than the inner diameter of the threaded insert **172** such that upon loosening of the fastener **151** while still engaged with the threaded insert **172**, the fastener **150**, washer **160** and control assemblies are not completely disassembled. Since the threads of the fastener **152** and threaded insert **172** remain engaged, simple tightening of the knob **171** results in closing of the

clamp assembly **110**. This loosely coupled arrangement of the clamp **100** further facilitates single handed operation.

(26) The lower **130** and upper **111** bodies may comprise indexing features **180** that facilitate one-handed operation of the clamp. When in operation the clamp **100** must be repositioned, or alternatively, when posts **104** must be inserted, removed or repositioned, it is desirable that, upon loosening of the fastener assembly **170**, only a small amount of vertical movement is allowed while the lower **130** and upper **111** bodies remain engaged. Referring to FIG. 3, this is achieved on the lower body **130** by two sets of indexing recesses **181a**, **181b** disposed in the lower front portion **133** and on the upper body **111** by spring loaded pins **182a**, **182b** disposed in holes **183a**, **183b** in the back portion **115** thereof. In operation, when the lower **130** and upper **111** bodies are in close proximity, pins **182a**, **182b** each engage one of a plurality of indexing recesses **181a**, **181b** thereby retaining the vertical alignment of the bodies **130**, **111**. For example, a vertical opening movement between the bodies **130**, **111** is possible by single hand squeezing of the top portion **131** of the lower body **130**, and the bottom **113** of the upper body **111**. Conversely, a vertical closing movement between the bodies **130**, **111** is possible by single hand squeezing of the top **112** of the upper body **111**, and the bottom **132** of the lower body **130**. Because of the indexing retention of the upper and lower bodies **130**, **111**, the hand may release the clamp assembly and move to the control assembly without the clamp falling off the rail **102**. This frees a second hand for another simultaneous task, an important advantage in, for example, a structure, support table. and/or in an operating room environment.

(27) Relative dimensions of the components that comprise clamp **100** may vary depending on the purpose served, for example, clamping the side rail **102** and clamping to the posts or pins **104**. However, clamp assembly **110**, fastener assembly **150**, washer assembly **160**, and control assembly **170** may be dimensioned differently for another application, such as clamping objects to the standing seam of a metal roof. In the first example, primary clamping is to be performed to operably connect the clamp body assembly **110** to the side rail **102** by the side rail arms **119** and **136** being pulled together by the tightening of the control assembly **170** advancing on the threaded portion **152** of fastener **151** and pressing the washer assembly **160** against the upper body **111**. Once the side rail **102** is compressed by the side rail arms **119**, **136**, e.g., in the vertical plane, advancing the knob **171** on the threads **152** moves the fastener assembly **150** in the openings of the channel portion **120** and key portion **140**, whereby the slotted fastener opening **123** of the upper body **111** causes the fastener **150** and control **170** assemblies to compress the back portion **115** of the upper body **111** towards the front portion **133** of the lower body **130** of the clamp assembly **110**. This action secures the posts **104** in the openings **117** (**117a**, **117b**), and the portions of posts **104** that protrude radially from the slot openings **118** (**118a**, **118b**) are engaged by the slot recesses **138** (**138a**, **138b**), thereby achieving a compressing effect in the horizontal plane. The directions in which clamping is effected may be mutually orthogonal, wherein compressive forces act in the horizontal and vertical direction, or alternatively compressing forces may act in non-orthogonal, but offset, directions, depending on the application.

(28) Referring to FIGS. 5-7E, a second embodiment for a clamp apparatus, system and method for a single action locking multiple components in different orientations is generally shown as element **100**. The clamp **100** has four component parts: a clamp assembly **110**, a fastener assembly **150**, a washer assembly **160**, and a control assembly **170**. The clamp assembly **110** comprises an upper body **111** configured to operatively connect to a lower body **130**. The upper body **111** comprises an upper front portion **114** and an upper side rail arm **119** with an upper side rail register **126b** for engaging a rail, tube or other elongated object. The lower body **130** comprises a lower front portion **133**, a lower back portion **134**, and a lower side rail arm **136** with a lower side rail register **126a** for engaging a rail, tube or other elongated object. Referring to FIG. 6B, the lower body **130** comprises an opening **117** for a post **104**, rod, tube or other cylindrical object to be fixedly attached by the clamp apparatus **100**. The lower body **130** also comprises a slot recess **138** configured to receive a

post **104**. The clamp assembly **110** can be assembled such that the slot recess **138** is located adjacent the slot opening **118** in the upper body **111**, best viewed in FIG. 6A, to accomplish single action locking of multiple components in different orientations. The slot recess **138** in the lower body **130** engages the portion of the post **104** that radially protrudes from the slot opening **118** in the upper body **111**. Optional pads **129a**, **129b** may be used to protect the surface finish of the rail or tube (not shown) and provide additional friction for clamp stability. The fastener assembly **150** comprises a T-segment **153** with cylindrical surface **157** and flat ends **156**. A threaded hole **154** is disposed in the cylindrical surface **157** approximately at its midpoint and is configured to receive a stem **158** with a threaded portion **159** orthogonally oriented to the T-segment **153**. A flat **159a** is disposed in the threaded portion **159** of the stem **158** at the end that is inserted into the threaded hole **154**. An axial, threaded hole **155** is disposed in the T-segment and opens into the threaded hole **154** disposed in the cylindrical surface **157**. The axial, threaded hole **155** is configured to receive a set screw **155a** that engages the flat **159a** in the threaded stem **158** and retains it in the fastener assembly **150**. Alternatively, the T-swivel configuration of the fastener assembly **150** may comprise a pin through a hole in the stem **158**, the pin inserted through an opening **146** in the upper body **111** simultaneously securing the stem **158** and plugging the opening **146**. The fastener assembly **150** is inserted into a fastener head receiver **144** disposed in the upper front portion **114**. The fastener head receiver **144** is coupled to an upper vertical slot **143** configured to accept the threaded stem **158**, **159** of the fastener assembly **150** thereby allowing limited rotational motion of the stem **158**. A dowel plug **145** may be used to fill any opening **146** left by the milling process. The fastener assembly **150** and upper body **111** mate to the lower body **130** with the threaded stem **158**, **159** passing through a fastener opening **123** having the shape of a lower vertical slot **124**, as shown in FIG. 5. The washer **160** and control **170** assemblies are disposed on the exposed, threaded end **159** of the stem **158** on the front side **133** of the lower body **130**. The washer assembly **160** comprises a washer body **161** with a centrally disposed opening **162** configured to allow the stem **158** to pass through, and an interior portion **165** configured to mate with the control assembly **170**. The control assembly **170** comprises a threaded knob **171** configured to receive a threaded end **159** of the stem **158**. It is to be appreciated by those of skill in the art that any armature that provides for the tightening and loosening by the threads would be suitable to use in may offer additional advantages. The washer assembly **160** further comprises a spherical surface **166** configured to mate with a plurality of bowl-shaped recesses **125** disposed in the lower front portion **133** of the lower body **130** and along the lower vertical slot **124** and allow for smooth angular displacement of the fastener assembly **150** by the tightening of the control assembly **170**. Alternatively, the surface of the washer may have a cylindrical shape with a plurality of matching cylindrical recesses disposed along the lower vertical slot **124** in the lower front portion **133** of the lower body **130**.

(29) The index clamping method **400** of the jogging clamp **100** may best be viewed in the sequence of operational steps outlined in FIGS. 7A-7E. In FIG. 7A, the jogging clamp **100** is shown in a first, clamped position wherein the control knob **171** is fully tightened on the stem **158**, **159** forcing the washer **161** into a first bowl-shaped recesses **125**. Tightening control assembly **170** forces the upper body **111** and the lower body **130**, including upper **119** and lower **136** side rail arms, toward one another to achieve clamping forces in multiple directions, simultaneously: First, a clamping force may be achieved in a direction that allows upper and lower bodies **111** and **130** to fixedly attach to, for example, rail **102**; second, a clamping force may be achieved in a direction that allows upper and lower bodies **111** and **130** to fixedly attach to, for example, a post **104**. As previously described, the fastener head receiver **144**, in conjunction with bowl-shaped recess **125** and spherical portion **166**, provide for angular movement that achieves simultaneous clamping of multiple components, by allowing the T-segment **151** and stem **158** to rotate within the lower vertical slot **124**.

(30) In a first index clamping step **401**, shown in FIG. 7B, the jogging clamp **100** is moved to a first, unclamped position wherein the control knob **171** is loosely threaded on the stem **158**, **159**

allowing the washer **161** to disengage from the first bowl-shaped recess **125**. A limited decoupling of the upper body **111** from the lower body **130** is possible in this configuration wherein the post **104** remains in the slot opening **118** of the upper body **111** and the through opening **117** of the lower body **130**, allowing the post to slide freely.

(31) In a second index clamping step **402**, shown in FIG. 7C, the jogging clamp **100** is shown in an open jogging position wherein the control knob **171** is loosely threaded on the stem **158**, and both upper **111** and lower **130** bodies may freely slide relative to each other along the post **104**. In this position, the upper body **111**, fastener **150**, washer, **160** and control **170** assemblies are translated together so as to accommodate a larger object to be inserted between upper **119** and lower **136** side rail arms.

(32) In a third index clamping step **403**, shown in FIG. 7D, the jogging clamp **100** is moved to a second, unclamped position wherein the control knob **171** is loosely threaded on the stem **158**, **159** allowing the washer **161** to align to and/or partially engage a second bowl-shaped recess **125**.

(33) In a fourth index clamping step **404**, shown in FIG. 7E, the jogging clamp **100** is closed around a rail, tube or other object (not shown). The control knob **171** is tightened on the stem **158**, **159** forcing the washer **161** into the second bowl-shaped recesses **125** thereby urging the upper **111** and lower **130** bodies, including upper **119** and lower **136** side rail arms, toward one another to achieve clamping on the rail and post **104** simultaneously.

(34) The index clamping method **400** may be varied by selecting alternate indexing positions, as defined by the bowl-shaped recesses **125** to accommodate larger or smaller primary clamping surfaces. The size of the simultaneously clamped post **104**, however, is defined by the diameter of the slot recess **117** and opening **118** and must be predetermined based on the application. The concepts disclosed herein with respect to the various embodiments are combinable, interchangeable, and non-limiting in that respect. For example, one skilled in the art will recognize that the jogging clamp **100** may be combined with certain embodiments described herein, as the intended purpose may serve.

(35) While certain configurations of structures have been illustrated for the purposes of presenting the basic structures of the present invention, one of ordinary skill in the art will appreciate that other variations are possible which would still fall within the scope of the appended claims. Additional advantages and modifications will readily occur to those skilled in the art. Therefore, the invention in its broader aspects is not limited to the specific details and representative embodiments shown and described herein. Accordingly, various modifications may be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents.

Claims

1. A clamp comprising: an upper body including: an upper base; an upper arm extending outwardly from said upper base; a channel portion formed in said upper base bottom portion configured to receive a key protrusion; and one or more slot-shaped fastener openings extending through said upper base; a lower body including: a lower base, a lower arm extending outwardly from said lower base, said upper and lower arms being adapted to clamp or couple to one or more first objects, one or more fastener head openings formed through said lower base, each of said fastener head openings having at least one rotational inhibitor; a key protrusion extending outwardly from said lower base, said key protrusion forming a complementary shape with said channel portion when said upper and lower bodies are in an assembled configuration, and a lower bottom portion, said lower bottom portion and said one or more openings being adapted to clamp or couple to one or more second objects; a fastener assembly comprising a bolt head and a threaded portion, said fastener assembly configured to be received and extend through each of said one or more fastener openings and said one or more fastener head openings, said bolt head having at least one

protrusion, said bolt head being configured to be received by said fastener head opening, said at least one rotational inhibitor and said at least one protrusion being adapted to prohibit rotation of the fastener assembly with respect to said upper and/or lower bodies, said fastener assembly further configured to translate along said one or more slot-shaped fastener openings to provide slidable adjustment of said threaded portion through said fastener opening; and a control assembly operably connected to said fastener assembly, said control assembly configured to advance along said threaded portion of said fastener assembly to urge said upper and lower body portions together to achieve simultaneous, multi-directional clamping of said one or more first and said one or more second objects.

2. The clamp according to claim 1, further comprising a washer assembly including a body and a washer opening extending therethrough and configured to receive said threaded portion of said fastener assembly.

3. The clamp according to claim 1, wherein said bolt head comprises six protrusions formed as a hex bolt, said one or more fastener head openings comprising a corresponding six notches.

4. A clamp comprising: an upper body including: an upper base; an upper arm extending outwardly from said upper base; a fastener head receiver formed in said upper base; an upper transverse opening formed along a portion of said upper base in said upper body intersecting said fastener head receiver; and an upper vertical slot formed in said upper front portion, said upper vertical slot intersecting said fastener head receiver; a lower body including: a lower base; a lower arm extending outwardly from said lower base, said upper and lower arms being adapted to clamp or couple to one or more first objects, a plurality of spherical recesses formed along said lower base in a lower front portion; and a lower vertical slot formed in said lower front portion intersecting said plurality of spherical recesses; a fastener assembly comprising a T-portion and a threaded portion, said threaded portion configured to be received and extend through said upper and lower transverse openings, said T-portion configured to be received and translationally fixed within said fastener head receiver, said fastener assembly further configured to rotate about said T-portion to provide angular adjustment of said threaded portion through said upper and lower vertical slots; and a control assembly operably coupled to said threaded portion of said fastener assembly, said control assembly configured to advance along said threaded portion to urge a portion of said control assembly into a spherical recess of said plurality of spherical recesses, thereby urging said upper and lower bodies together to achieve simultaneous, multi-directional clamping of said first and second objects.

5. The clamp according to claim 4, wherein said portion of said control assembly comprises a washer forming a complementary shape as that of one spherical recess of said plurality of spherical recesses.

6. A method of adjusting a clamp for clamping to variable-sized first objects, the method comprising: providing the clamp according to claim 4; loosening said control assembly so as to disengage said control assembly from a first spherical recess of said plurality of spherical recesses; rotating the control assembly about said T-portion so that said an upper body is slidably moveable relative to said lower body; aligning said control assembly with a second spherical recess of said plurality of spherical recesses; and tightening said control assembly so as to engage said control assembly with said second spherical recess, thereby moving said upper and lower bodies, thereby effecting a clamping action.

7. The method according to claim 6, wherein said portion of said control assembly comprises a washer forming a complementary shape as that of one spherical recess of said plurality of spherical recesses.
